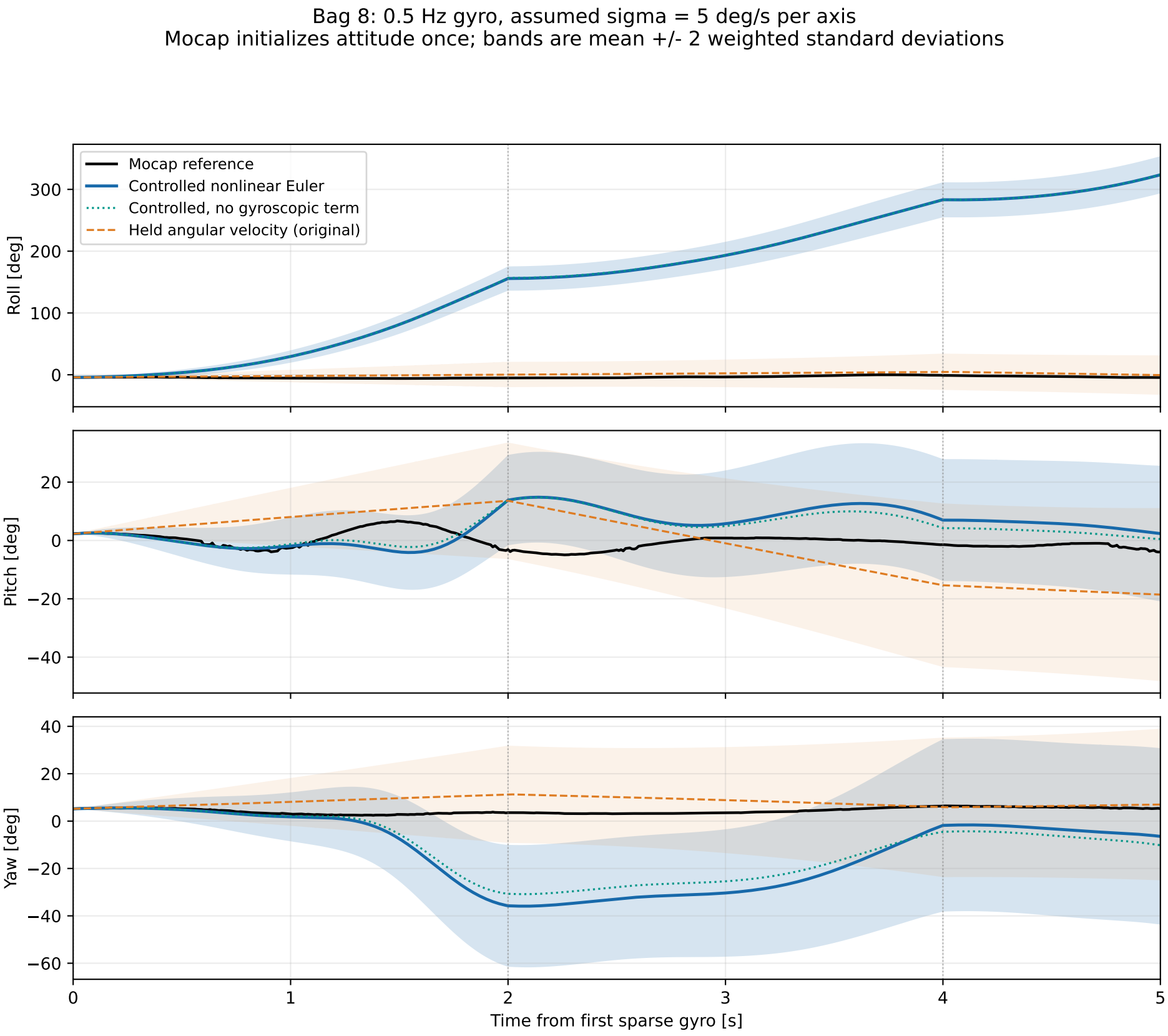
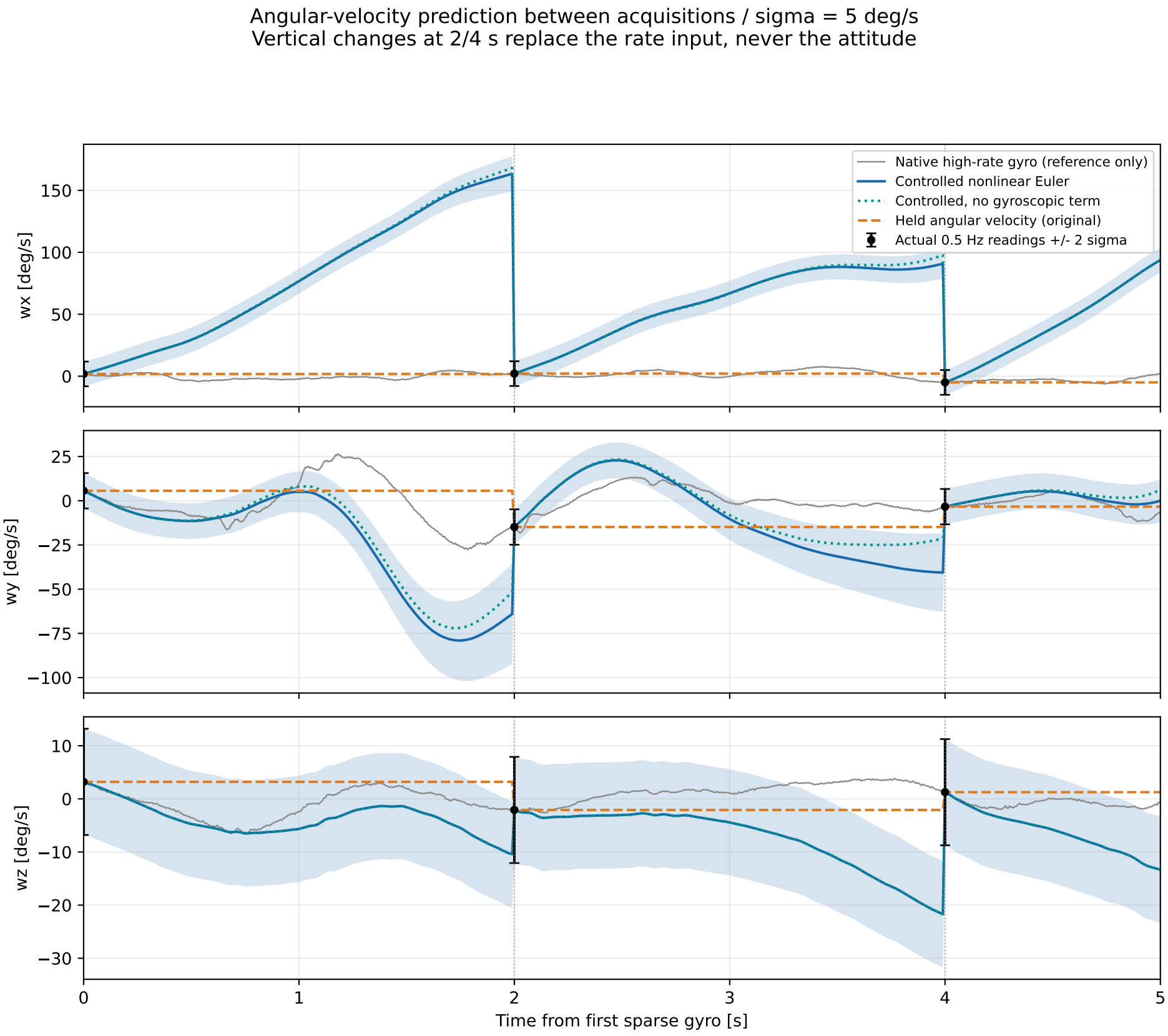
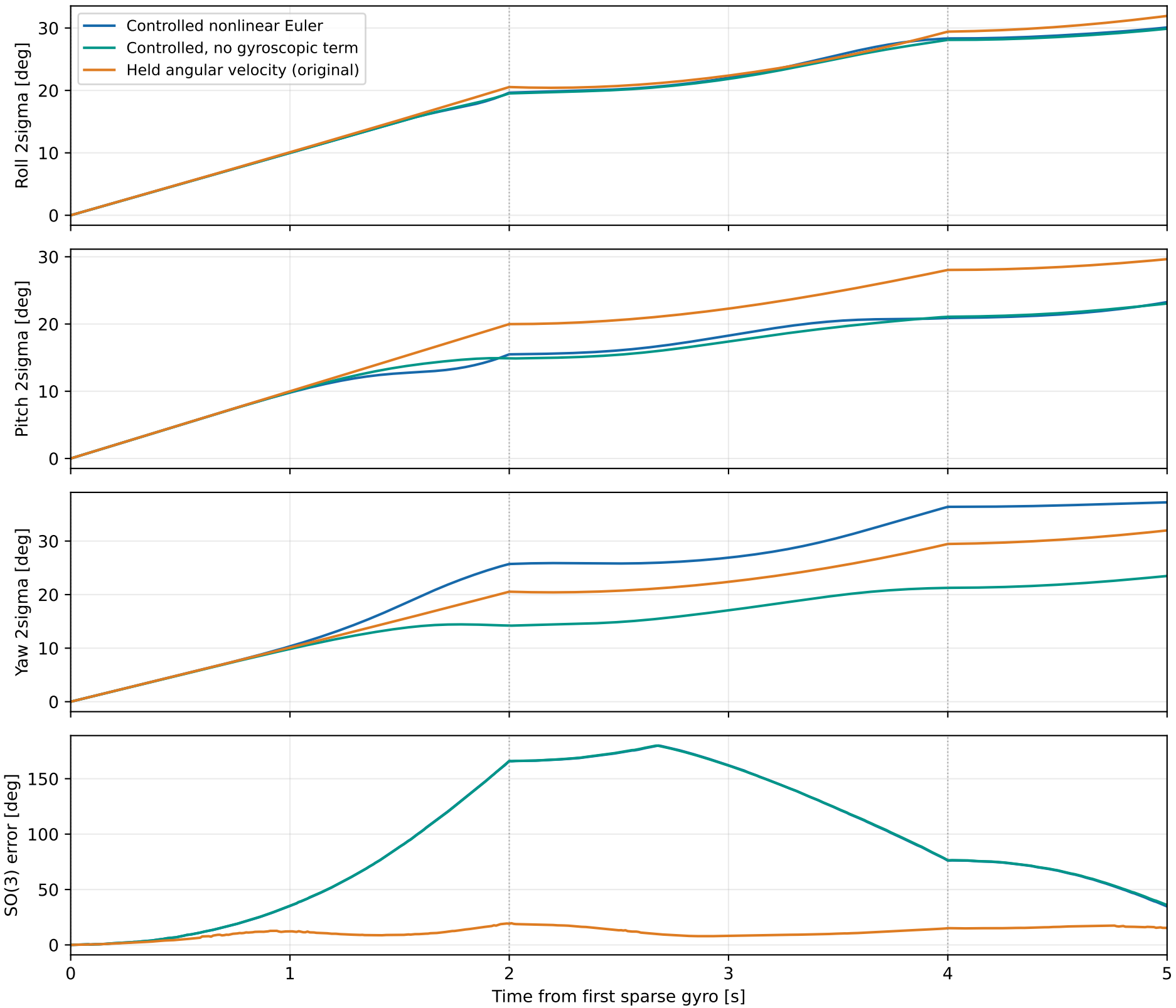

Bag 8: 0.5 Hz gyro, assumed sigma = 5 deg/s per axis
Mocap initializes attitude once; bands are mean +/- 2 weighted standard deviations
Mocap reference
Controlled nonlinear Euler
Controlled, no gyroscopic term
Held angular velocity (original)
Roll [deg]
Pitch [deg]
Yaw [deg]
Time from first sparse gyro [s]


Angular-velocity prediction between acquisitions / sigma = 5 deg/s
Vertical changes at 2/4 s replace the rate input, never the attitude
Native high-rate gyro (reference only)
Controlled nonlinear Euler
Controlled, no gyroscopic term
Held angular velocity (original)
Actual 0.5 Hz readings +/- 2 sigma
wx [deg/s]
wy [deg/s]
wz [deg/s]
Time from first sparse gyro [s]


Propagated spread and mocap comparison / sigma = 5 deg/s
Conditional on fixed nominal inertia, actuator inputs and past residual-torque mean
Controlled nonlinear Euler
Controlled, no gyroscopic term
Held angular velocity (original)
Roll 2sigma [deg]
Pitch 2sigma [deg]
Yaw 2sigma [deg]
SO(3) error [deg]
0
10
20
30
0
50
100
150
0
1
2
3
4
5
Time from first sparse gyro [s]